US 20250282220A1

(19) United States
(12) Patent Application Publication
Nash et al.

(10) Pub. No.: US 2025/0282220 A1
(43) Pub. Date: Sep. 11, 2025

(54) VEHICLE DISPLAY CONTROL

(71) Applicant: **Ford Global Technologies, LLC**, Dearborn, MI (US)

(72) Inventors: **Brian Nash**, Plymouth, MI (US); **Kevin Lee Helpingstine**, Ferndale, MI (US); **Brendan Francis Diamond**, Grosse Pointe, MI (US)

(73) Assignee: **Ford Global Technologies, LLC**, Dearborn, MI (US)

(21) Appl. No.: **18/599,537**

(22) Filed: **Mar. 8, 2024**

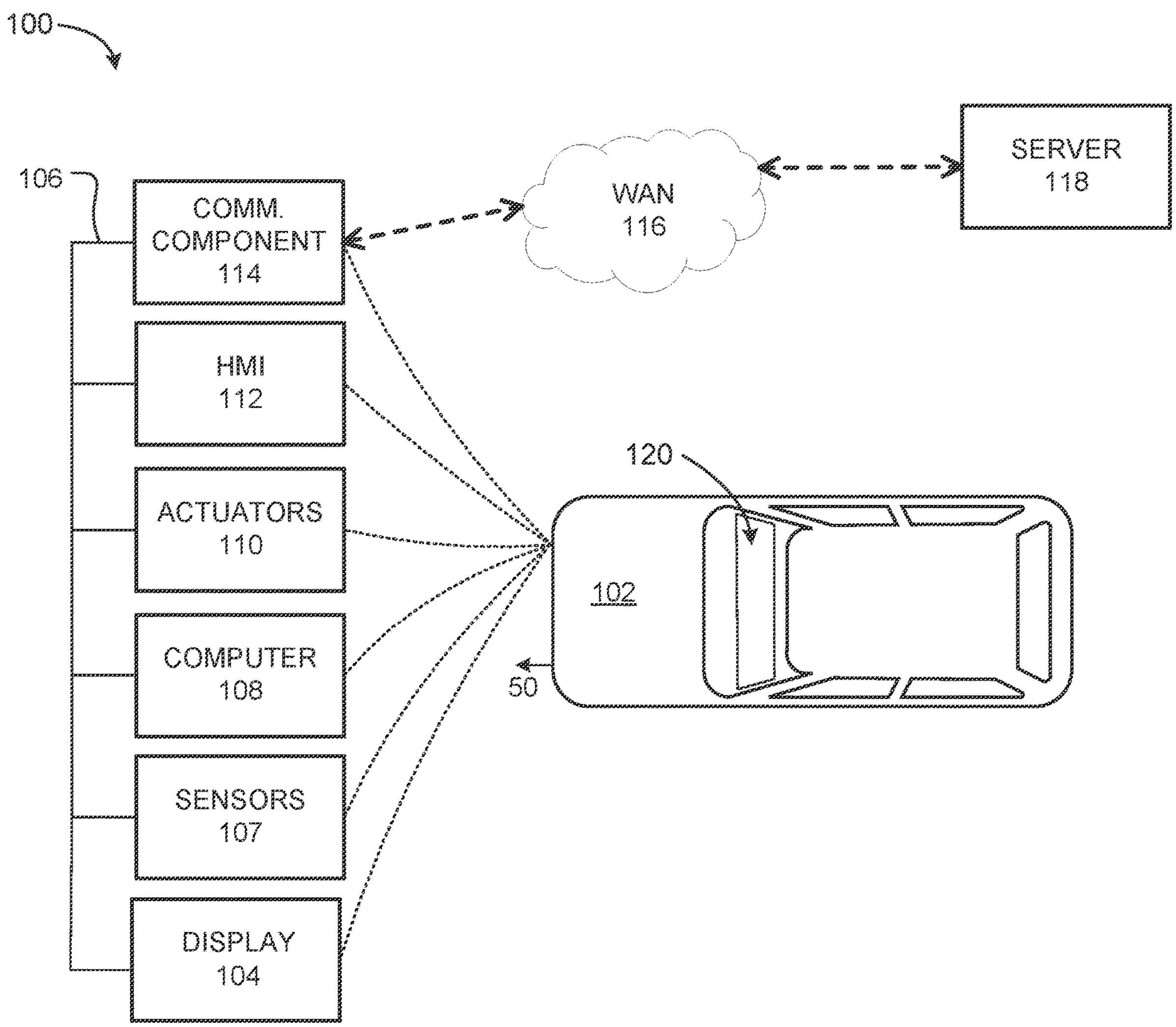
Publication Classification

(51) Int. Cl.
B60K 35/28 (2024.01)
H04N 7/15 (2006.01)

(52) U.S. Cl.
CPC ***B60K 35/28*** (2024.01); ***H04N 7/15*** (2013.01); *B60K 2360/195* (2024.01); *B60K 2360/741* (2024.01)

(57) ABSTRACT

A computer includes a processor and memory, the memory can store instructions executable by the processor to transition a vehicle display from a vehicle monitored state to a content delivery state based on a placement of the vehicle in a non-monitored state, and to transition a control from a vehicle control state to a content delivery control state based on the placement of the vehicle in the non-monitored state.



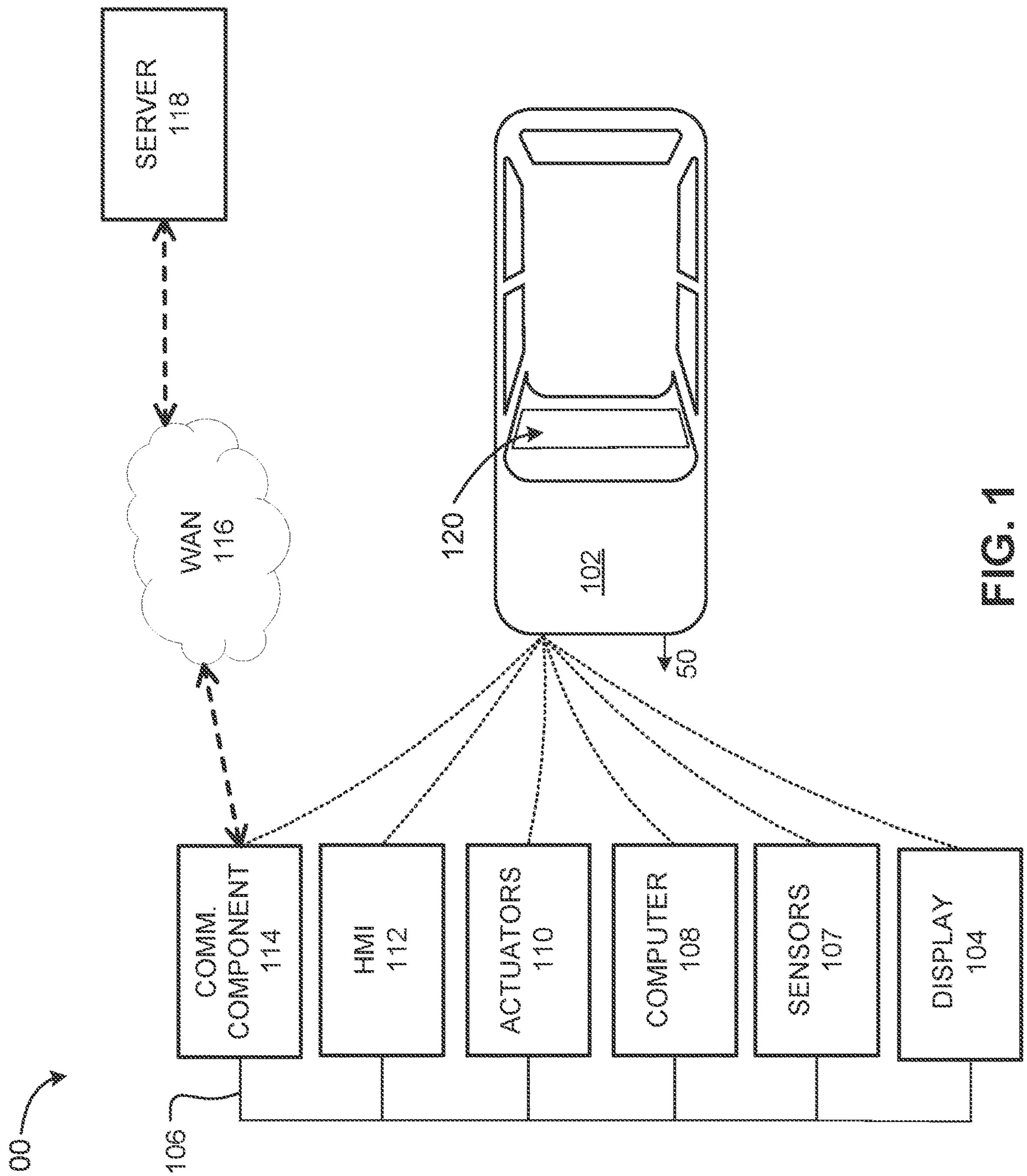

SERVER
118
WAN
116
120
102
50
COMM.
COMPONENT
114
HMI
112
ACTUATORS
110
COMPUTER
108
SENSORS
107
DISPLAY
104
106
100

FIG. 1

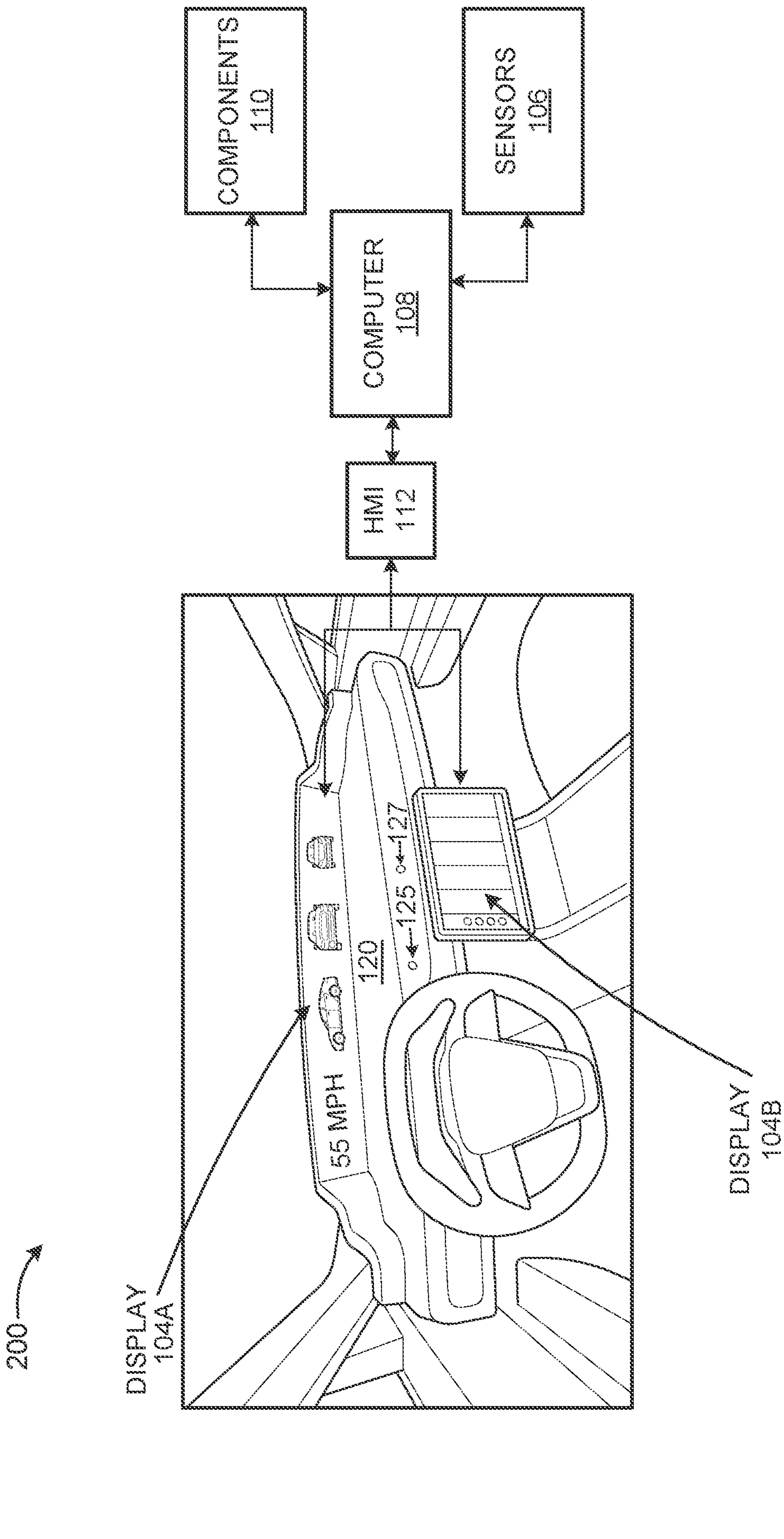

COMPONENTS
110
SENSORS
106
COMPUTER
108
HMI
112
55 MPH
120
125
127
DISPLAY
104A
DISPLAY
104B
200

FIG. 2

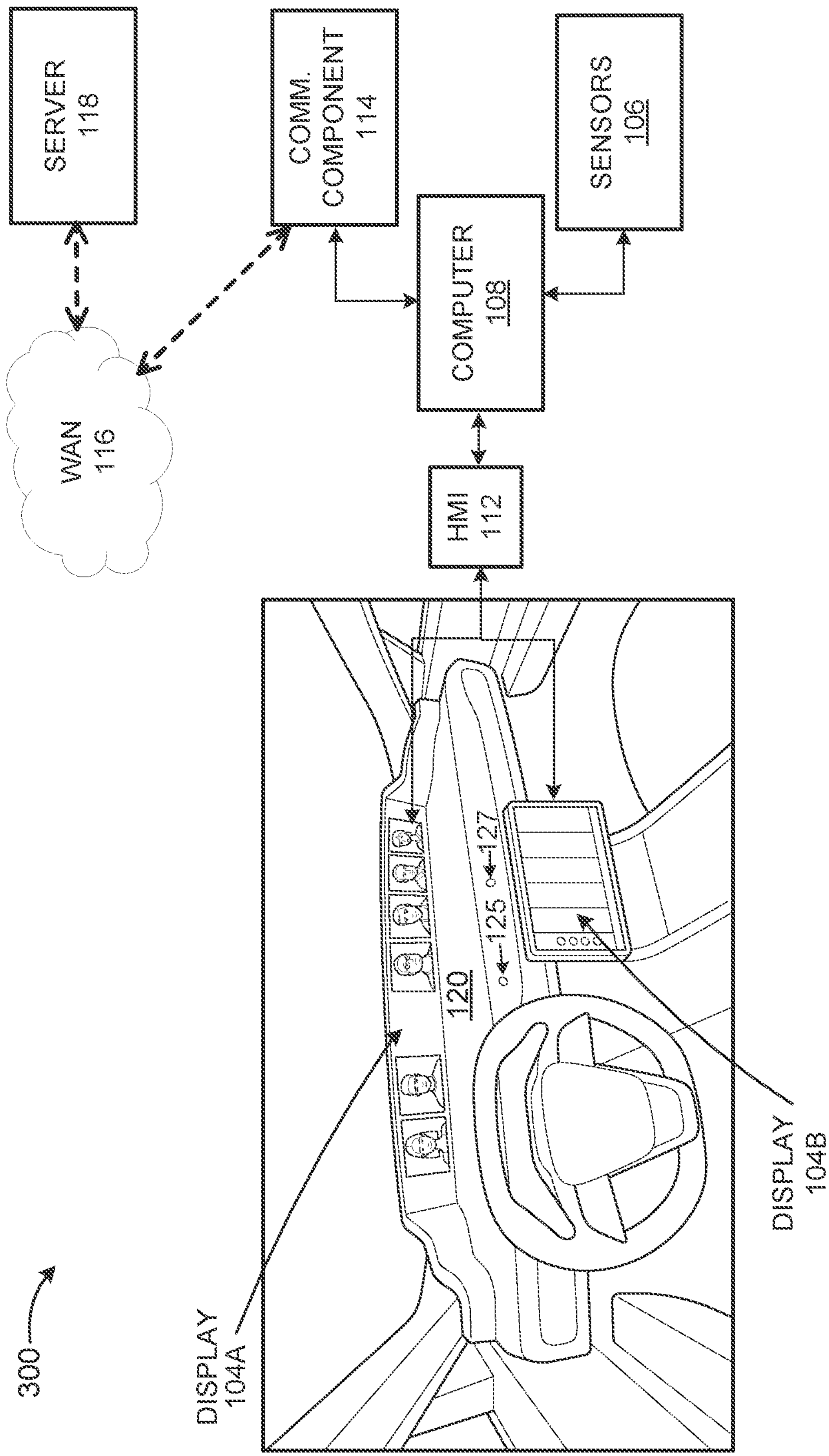

SERVER
118
COMM.
COMPONENT
114
SENSORS
106
COMPUTER
108
WAN
116
HMI
112
120
125
127
DISPLAY
104A
DISPLAY
104B
300

FIG. 3

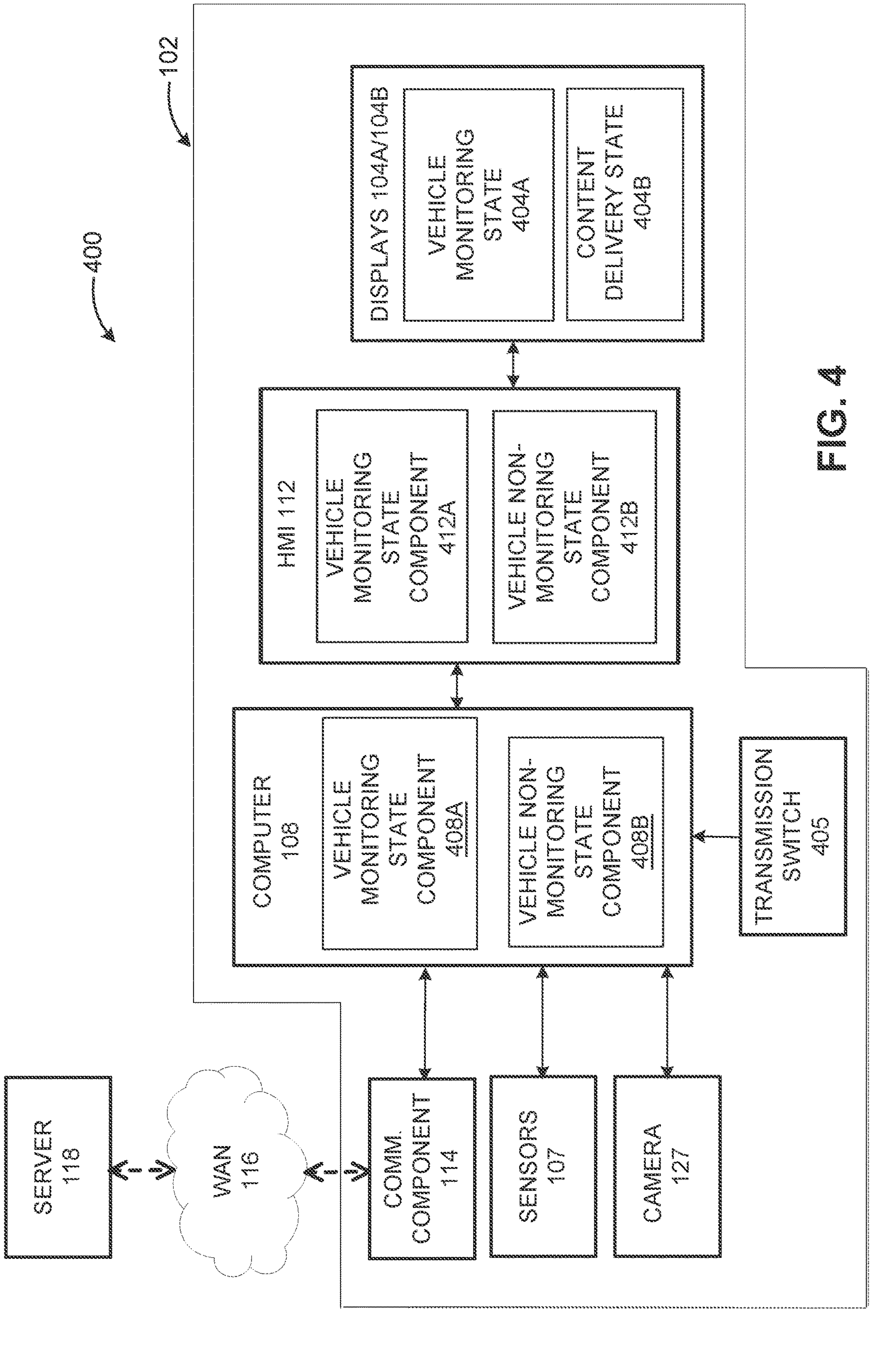

400
102
DISPLAYS 104A/104B
VEHICLE MONITORING STATE 404A
CONTENT DELIVERY STATE 404B
HMI 112
VEHICLE MONITORING STATE COMPONENT 412A
VEHICLE NON-MONITORING STATE COMPONENT 412B
COMPUTER 108
VEHICLE MONITORING STATE COMPONENT 408A
VEHICLE NON-MONITORING STATE COMPONENT 408B
TRANSMISSION SWITCH 405
SERVER 118
WAN 116
COMM. COMPONENT 114
SENSORS 107
CAMERA 127

FIG. 4

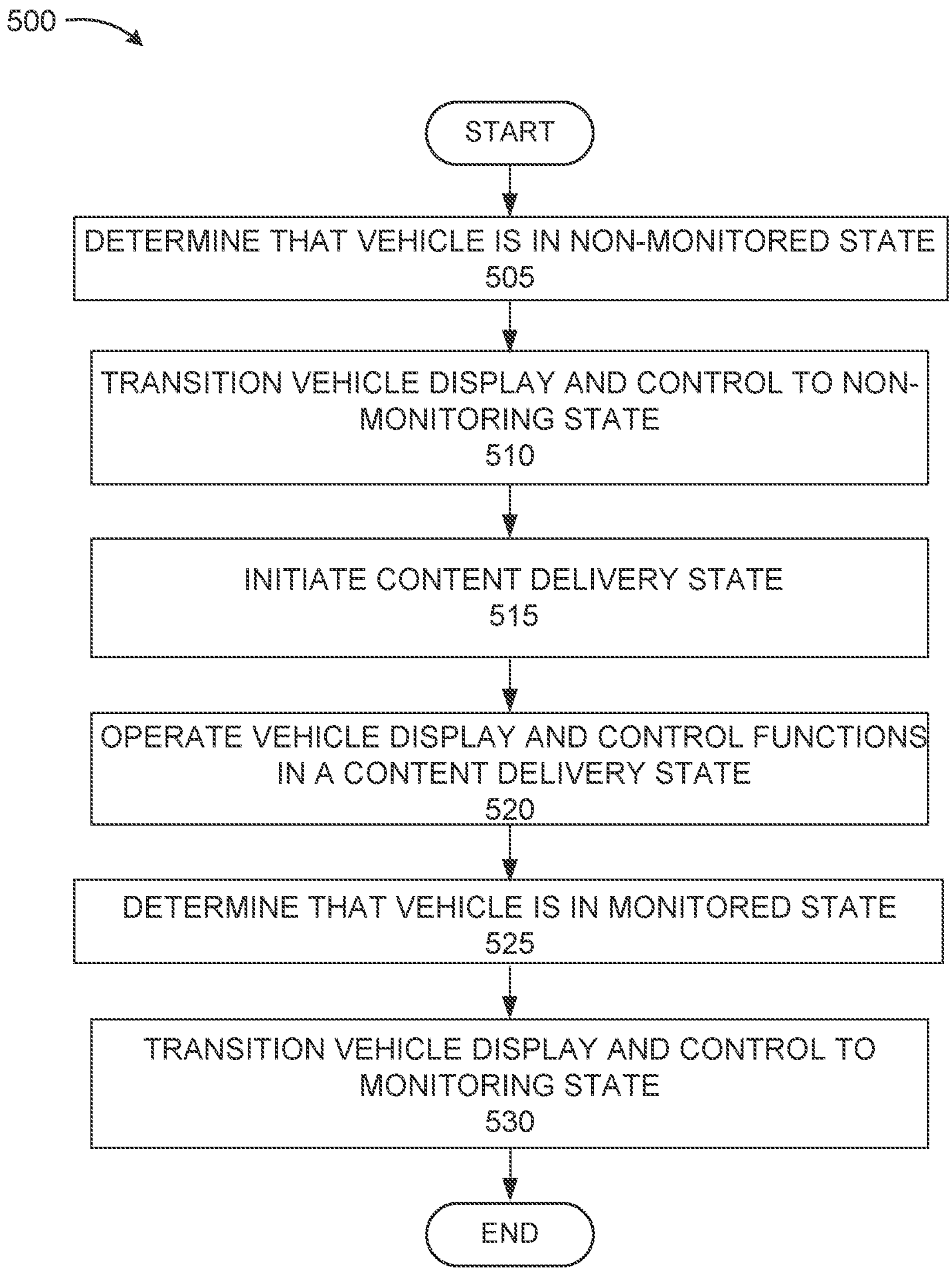

500
START
DETERMINE THAT VEHICLE IS IN NON-MONITORED STATE
505
TRANSITION VEHICLE DISPLAY AND CONTROL TO NON-MONITORING STATE
510
INITIATE CONTENT DELIVERY STATE
515
OPERATE VEHICLE DISPLAY AND CONTROL FUNCTIONS IN A CONTENT DELIVERY STATE
520
DETERMINE THAT VEHICLE IS IN MONITORED STATE
525
TRANSITION VEHICLE DISPLAY AND CONTROL TO MONITORING STATE
530
END

FIG. 5

VEHICLE DISPLAY CONTROL

BACKGROUND

[0001] A vehicle may include a monitoring system that includes one or more displays, indicators, and/or other devices capable of providing information to an operator. Vehicle monitoring system information may relate to an environment external to the vehicle, which may include relative positions of static or moving objects in a traffic environment, road markings and boundaries, buildings, bicycles, etc. A vehicle monitoring system may additionally provide information relating to components internal to a vehicle, such as engine operating parameters, battery state parameters, component operating temperatures, fluid pressures, etc.

BRIEF DESCRIPTION OF THE DRAWINGS

[0002] FIG. **1** is a block diagram of an example vehicle system.

[0003] FIGS. **2** and **3** are diagrams showing displays positioned in an interior portion of a vehicle.

[0004] FIG. **4** is a block diagram showing vehicle operation of a vehicle system in a vehicle monitored state and in a vehicle non-monitored state.

[0005] FIG. **5** is a flowchart of an example method for transitioning between a vehicle monitored state and a vehicle non-monitored state.

DESCRIPTION

[0006] During operation of a vehicle, a vehicle monitoring system may monitor and display information relating to a traffic environment, such as positions of static or moving objects relative to a vehicle. Such objects can include static or moving vehicles, bicycles, road markings, signposts, lampposts, buildings, etc. Such information, which may be gathered utilizing an externally-facing camera mounted on the vehicle, may assist an operator in obtaining and maintaining situational awareness of the traffic environment. Display devices of a vehicle monitoring system may be positioned on or in a vehicle dashboard beneath a lower boundary of the vehicle windshield so as to be viewable by the vehicle operator while the operator is focused on an object viewable through the vehicle windshield. In this context, a "display" means a physical device having a screen (such as a screen liquid-crystal-based-screen, light-emitting-diode-based screen, a plasma-based screen, etc.) that presents information in a visual form. In an example, displays of a vehicle monitoring system can be lengthened to extend across the vehicle dashboard so as to provide a panoramic display that extends from a left internal boundary of a forward portion of a vehicle interior to a right internal boundary of the forward portion of the vehicle interior, e.g., from side-to-side. Accordingly, an operator may be able to view at least a portion of the panoramic display while focusing on an object viewable through the vehicle windshield.

[0007] A vehicle monitoring system may monitor operations internal to the vehicle, such as operations relating to mechanical components, actuators, other vehicle systems and subsystems, etc. Thus, while operating a vehicle, a vehicle operator may maintain awareness of vehicle engine operating parameters, fuel and/or battery status, climate control settings, cruise control settings, radio settings, etc. Accordingly, a vehicle monitoring system, which may include a panoramic display, may assist in providing an operator with an overall driving experience that includes a capability for monitoring an environment external to the vehicle as well as monitoring an environment internal to the vehicle.

[0008] An overall driving experience can be enhanced by a facilitating capability for an operator to communicate wirelessly with others, such as by way of cellular communications that interoperate with a vehicle audio system, e.g., microphones, speakers, etc., operating within the vehicle's interior. As described in reference to examples herein, in response to a vehicle operator receiving a request for a videoconference, a display and a computing resource of a vehicle monitoring system can be utilized to permit the operator to participate in a videoconference. In an example as described further herein, in response to receiving a request for a videoconference, the operator may transition the vehicle monitoring system from a monitored state, which, in this context, means a vehicle monitoring system utilized to monitor a traffic environment and/or an environment internal to the vehicle, to a non-monitored state, which, in this context, means a vehicle monitoring system that is not utilized to monitor a traffic environment or environment internal to the vehicle. Also in this context, the term "videoconference" means a conference in which participants in different locations are able to communicate with each other in sound and/or vision. Accordingly, a videoconference may include multiparty audio telephony and/or video telephony or any other technique in which audio and/or video streams from a plurality of remote stations is assembled and presented at a destination receiver.

[0009] A vehicle monitoring system may be transitioned to a non-monitored state via an operator bringing the vehicle to an at-rest state, which, in this context, means exiting a traffic environment and placing the vehicle's transmission into a state that prevents movement of the wheels of the vehicle. Based on an onboard vehicle computer, e.g., computer **108** of FIG. **1**, determining that the vehicle has been placed in an at-rest state, the vehicle monitoring system may be transitioned from a vehicle monitored state to a vehicle non-monitored state. In a vehicle non-monitored state, images from a camera mounted on an externally-facing surface of the vehicle may be suppressed and replaced by content delivered during a videoconferencing session. Accordingly, in this context, a "content delivery" state means an operating state in which images relating to the vehicle's external environment and/or the vehicle's internal environment are suppressed and replaced by videoconferencing content. In an example, in a content delivery state, images (e.g., real-time images) of videoconference participants, as well as content provided by a videoconference participant (e.g., view graphs that include still or video images) may be displayed in place of images of a vehicle's external environment and/or images relating to a vehicle's internal operating state. In an example, audio from videoconference participants can be presented to the operator via the vehicle's interior audio system.

[0010] Thus, in a content delivery state, an operator may participate in a videoconference utilizing vehicle displays viewable from the vehicle's interior. A vehicle interior camera, e.g., a dashboard camera, may be utilized to capture images of the operator as a videoconference participant. A vehicle interior audio system, e.g., a speaker, a microphone, etc., may be utilized to communicate audio signals between the vehicle operator and other videoconference participants. In an example, use of a panoramic display, which extends from a first location proximate to a left interior boundary to a second location proximate to a right interior boundary, can be populated with images from some or all videoconference participants and/or materials presented (e.g., view graphs, audio and/or video segments, etc.) during a videoconference. In an example, control elements, such as elements used to control radio settings, cruise control settings, etc., can be transitioned from a vehicle monitoring state to a vehicle non-monitoring state, e.g., to control videoconferencing features. Thus, in an example, haptic or tactile actuators, such as an actuator positioned in a driver side seat cushion, an actuator positioned in the vehicle steering wheel, etc., can be utilized to call the operator's awareness to certain content, e.g., posting of a videoconferencing chat message, which may have increased relevance to the operator. In another example, in a vehicle non-monitoring state, a haptic or tactile actuator may provide an indication that an operator has been mentioned, e.g., by name, and is being requested by a videoconference participant to maintain awareness of a particular portion of a videoconferencing session.

[0011] In an example, a system can include a computer having a processor and memory, the memory storing instructions executable by the processor to transition a vehicle display from a vehicle monitoring state to a content delivery state based on a placement of the vehicle in a non-monitored state. The system can also include instructions to transition a control from a vehicle control state to a content delivery control state based on the placement of the vehicle in the non-monitored state.

[0012] In an example, the non-monitored state can include a state of a vehicle transmission switch.

[0013] In an example, the display can extend across a dashboard of the vehicle from a first location proximate to a left interior boundary to a second location proximate to a right interior boundary.

[0014] In an example, the instructions to transition the vehicle display can include instructions to suppress display of images that represent an environment external to the vehicle.

[0015] In an example, the instructions to transition the vehicle display can include instructions to suppress display of operating parameters of a vehicle system.

[0016] In an example, the instructions to transition the control from the vehicle control state can include instructions to filter audio signals emanating from an interior portion of the vehicle to include audio exclusively emanating from an operator of the vehicle or from a predetermined occupant of the vehicle.

[0017] In an example, the content delivery state is a videoconferencing state that can permit display of content transmitted from a source external to the vehicle.

[0018] In an example, the instructions to transition the control from the vehicle control state can include instructions to modify a volume setting for audio directed to a videoconferencing participant.

[0019] In an example, the instructions to transition the control from the vehicle control state can include instructions to detect reorientation of head or facial features of an operator of the vehicle.

[0020] In an example, the instructions to transition the control from the vehicle control state can include instructions to generate an audio output responsive to determining that videoconferencing content pertains to an operator of the vehicle.

[0021] In an example, the instructions to transition the control from the vehicle control state can include instructions to generate a haptic output to an operator of the vehicle responsive to determining audio or video content that pertains to the operator of the vehicle.

[0022] In an example, the instructions to transition the control from the vehicle control state can include instructions to suppress video signals from an operator of the vehicle based on detecting that the operator of the vehicle has become disengaged from the videoconference.

[0023] In an example, the instructions to transition the control from the vehicle control state can include instructions to suppress a video portion of a videoconference responsive to a determination that the vehicle has been placed in a monitored state.

[0024] In an example, the instructions to transition the control from the vehicle control state can include instructions to maintain an audio portion of a videoconference responsive to a determination that the vehicle has been placed in a monitored state.

[0025] In an example, a method can include transitioning a vehicle display from a vehicle monitored state to a content delivery state based on a placement of the vehicle in a non-monitored state. The method can additionally include transitioning a control from a vehicle control state to a content delivery control state based on the placement of the vehicle in the non-monitored state.

[0026] In an example, determining that the vehicle has been placed in the non-monitored state can include determining a state of a vehicle transmission switch.

[0027] In an example, the vehicle display can extend across a dashboard of the vehicle from a first location proximate to a left interior boundary to a second location proximate to a right interior boundary.

[0028] In an example, transitioning the vehicle display can include suppressing display of images representing an environment external to the vehicle.

[0029] In an example, transitioning the vehicle display can include suppressing display of operating parameters of a vehicle system.

[0030] In an example, the content delivery state is a videoconferencing state, in which the videoconferencing state permits display of content transmitted from a source external to the vehicle.

[0031] As shown in FIG. **1**, a system **100** includes a vehicle **102**, that includes computer **108** that is communicatively coupled, via vehicle network **106**, with various elements including sensors **107**, actuators **110**, such as steering, propulsion, braking, etc., human machine interface (HMI) **112**, and communication component **114**. Computer **108**, and server **118**, discussed below, include a processor and a memory. Memories of computer **108** and server **118**, such as those described herein, include one or more forms of non-transitory media readable by computer **108**, and can store first instructions executable by computer **108** for performing various operations, such that the vehicle computer is configured to perform the various operations, including those disclosed herein.

[0032] For example, computer **108** can include a generic computer with a processor and memory as described above and/or may comprise an electronic control unit (ECU) or a controller for a specific function or set of functions, and/or a dedicated electronic circuit including an ASIC (application specific integrated circuit) that is manufactured for a particular operation, (e.g., an ASIC for processing data from sensors and/or communicating data from sensors **107**). In another example, computer **108** can include an FPGA (Field-Programmable Gate Array), which is an integrated circuit manufactured to be configurable by a user. In example embodiments, a hardware description language such as VHDL (Very High Speed Integrated Circuit Hardware Description Language) may be used in electronic design automation to describe digital and mixed-signal systems such as FPGA and ASIC. For example, an ASIC is manufactured based on VHDL programming provided pre-manufacturing, whereas logical components inside an FPGA may be configured based on VHDL programming, e.g., stored in a memory electrically connected or coupled to the FPGA circuit.) In some examples, a combination of processor(s), ASIC(s), and/or FPGA circuits may be included in computer **108**. Further, computer **108** can include a plurality of computers in the vehicle (e.g., a plurality of ECUs or the like) operating together to perform operations ascribed herein to the computer **108**.

[0033] The memory of computer **108** can be of any type, such as a memory that utilizes hard disk drives, solid state drives, or any volatile or non-volatile media. The memory can store the collected data transmitted by sensors **107**. The memory can be a separate device from computer **108**, and computer **108** can retrieve information stored by the memory via a communication network in the vehicle such as vehicle network **106**, e.g., over a controller area network (CAN) bus, a local interconnect network (LIN) bus, a wireless network, etc. Alternatively or additionally, the memory can be part of computer **108**, for example, as a memory internal to computer **108**.

[0034] Computer **108** can include or access instructions to operate one or more actuators **110** such as actuators connected to the brakes of vehicle **102**, propulsion (e.g., one or more of an internal combustion engine, electric motor, hybrid engine, etc.), steering, climate control, interior and/or exterior lights, infotainment, navigation etc., as well as to determine whether and when computer **108**, as opposed to a human operator, is to control such operations. Computer **108** can include or be communicatively coupled, e.g., via vehicle network **106**, to more than one processor, which can be included in actuators **110**, such as sensors **107**, which may be utilized in vehicle electronic control units (ECUs) or the like included in the vehicle for monitoring and/or controlling various vehicle electromechanical components, e.g., a powertrain controller, a brake controller, a steering controller, etc.

[0035] Computer **108** can be generally arranged for communications on vehicle network **106** that can include a communications bus in the vehicle, such as a controller area network CAN or the like, and/or other wired and/or wireless mechanisms. Vehicle network **106** corresponds to a communications network, which can facilitate exchange of messages between various onboard vehicle devices, e.g., sensors **107**, actuators **110**, computer **108** and other computing resources onboard vehicle **102**. Computer **108** can be generally programmed to send and/or receive, via vehicle network **106**, messages to and/or from other devices in vehicle **102**, e.g., any or all of ECUs, sensors **107**, actuators **110**, communications component **114**, human machine interface (HMI) **112**, display **104**, etc.

[0036] Further, in implementations in which computer **108** actually comprises a plurality of devices, vehicle network **106** may be used for communications between devices represented as computer **108** in this disclosure. For example, vehicle network **106** can provide a communications capability via a wired bus, such as a CAN bus, a LIN bus, or can utilize any type of wireless communications capability. Vehicle network **106** can include a network in which messages are conveyed using any other wired communication technologies and/or wireless communication technologies, e.g., Ethernet, Wi-Fi®, Bluetooth®, etc. Additional examples of protocols that may be used for communications over vehicle network **106** in some implementations include, without limitation, Media Oriented System Transport (MOST), Time-Triggered Protocol (TTP), and FlexRay. In some implementations, vehicle network **106** can represent a combination of multiple networks, possibly of different types, that support communications among devices onboard a vehicle. For example, vehicle network **106** can include a CAN bus, in which some in-vehicle sensors and/or components communicate via a CAN bus, and a wired or wireless local area network in which some device in vehicle communicate according to Ethernet, Wi-Fi®, and/or Bluetooth communication protocols.

[0037] Vehicle **102** typically includes a variety of sensors **107**. Sensors **107** include a devices that can obtain one or more measurements of one or more physical phenomena. Some of sensors **107** can assist computer **108** in monitoring the environment external to vehicle **102**. For example, some of sensors **107** can include an externally facing camera, which may generate and transmit images to computer **108**. Instructions executed by computer **108** can be combined with output signals from other sensors of sensors **107**, such as radar sensors, lidar sensors, etc., which can provide a distance to, and/or a speed of, objects within the field of view of externally facing camera. In an example, instructions executed by computer **108** can generate symbols or other graphics that provide information relevant to a detected object, such as direction of movement of an object, speed of an object, distance to an object, etc.

[0038] Various sensors **107** may assist computer **108** in monitoring the operating environment internal to vehicle **102**, such as speed sensors, torque sensors, braking sensors, temperature sensors, etc., which operate to monitor and/or control vehicle speed settings, vehicle towing parameters, vehicle braking parameters, engine torque output, engine and transmission temperatures, battery temperatures, vehicle steering parameters, etc. Various sensors **107** may assist in characterizing the physical environment of vehicle **102**, such as outside air temperature, humidity, weather conditions (e.g., rain, snow, etc.), parameters related to the inclination or gradient of a road or other type of path on which the vehicle is proceeding, a particular ambient temperature, surface roughness of a road (e.g., on road versus off-road travel), etc. In example embodiments, sensors **107** can operate to detect the position or orientation of the vehicle utilizing, for example, signals from a satellite positioning system (e.g., global positioning system or GPS); accelerometers, such as piezo-electric or microelectromechanical systems MEMS; gyroscopes such as rate, ring laser, or fiber-optic gyroscopes; inertial measurement units IMU; and magnetometers.

[0039] Computer **108** can be configured for utilizing vehicle-to-vehicle (V2V) communications via communication component **114** and/or may interface with devices outside of the vehicle, e.g., through wide area network (WAN) **116** via V2V communications. Computer **108** can communicate outside of vehicle **102**, such as via vehicle-to-infrastructure (V2I) communications, vehicle-to-everything (V2X) communications, or V2X including cellular communications C-V2X, and/or wireless communications cellular dedicated short range communications DSRC, etc. Communications outside of vehicle **102** can be facilitated by direct radio frequency communications and/or via server **118**. Communications component **114** can include one or more mechanisms by which computer **108** communicates with computing entities outside of vehicle **102**, including any desired combination of wireless, e.g., cellular, wireless, satellite, microwave, radio frequency communication mechanisms and any desired network topology or topologies when a plurality of communication mechanisms are utilized.

[0040] Vehicle **102** can include HMI **112** (human-machine interface), e.g., one or more of an infotainment display, a touchscreen display, a microphone, a speaker, etc. A user, such as the operator of vehicle **102**, can provide input to devices such as computer **108** via HMI **112**. HMI **112** can communicate with computer **108** via vehicle network **106**, e.g., HMI **112** can send a message including the user input provided via a touchscreen of display **104**, microphone **125**, a camera that captures a gesture, etc., to computer **108**, and/or can display output, e.g., via display **104**, speaker, etc. In an example, operations of the HMI **112** can be included via instructions executed by computer **108**. Alternatively or additionally, the HMI **112** can include a computing device to execute and/or control operations ascribed herein to the HMI **112**.

[0041] During operation of vehicle **102** and a monitored state, HMI **112** can implement volume controls to control an onboard infotainment system, satellite radio system, etc., as well as providing controls for other onboard features of vehicle **102**. HMI **112** can additionally generate haptic outputs to an operator, such as activating a haptic actuator located in the driver side seat or on the steering wheel. In an example, a haptic actuator may output a predetermined series of vibrations, e.g., a vibration at a specified intensity for a specified period of time, which may notify the operator of vehicle **102** of an approaching (second) vehicle. In another example, a haptic actuator may output of predetermined series of vibrations to notify the operator of vehicle **102** of upcoming traffic congestion along path **50**. In another example, a haptic actuator located in the driver side seat of vehicle **102** may notify an operator of an incoming cellular telephone call.

[0042] WAN **116** can include one or more mechanisms by which computer **108** can communicate with server **118**. Server **118** can include an apparatus having one or more computing devices, e.g., having respective processors and memories and/or associated data stores, which may be accessible via WAN **116**. In example embodiments, vehicle **102** could include a wireless transceiver (i.e., transmitter and/or receiver) to send messages to (e.g., text, audio, still or video images, etc.) and receive messages from computing resources located outside of vehicle **102**. Accordingly, WAN **116** can include one or more of various wired or wireless communication mechanisms, including any desired combination of wired e.g., cable and fiber and/or wireless, e.g., cellular, wireless, satellite, microwave, and radio frequency communication mechanisms and any desired network topology or topologies when multiple communication mechanisms are utilized. Exemplary communication networks include wireless communication networks, e.g., using Bluetooth, Bluetooth Low Energy BLE, IEEE 802.11, V2V or V2X such as cellular V2X CV2X, DSRC, etc., local area networks and/or wide area networks, including the Internet.

[0043] In an example, computer **108** can utilize one or more sensors **107** to determine a transition of vehicle **102** from a vehicle monitored state, such as during operation of vehicle **102** in a traffic environment, to a vehicle non-monitored state, such as responsive to vehicle **102** exiting a traffic environment and placing the transmission of vehicle **102** in "park," or any other state that represents a state in which the transmission of vehicle **102** prevents movement of the wheels of the vehicle. Based on vehicle **102** being transitioned from a vehicle monitored state to a vehicle non-monitored state, instructions executed by computer **108** can suppress images of, for example, the environment external to vehicle **102**. In addition, based on transitioning of vehicle **102** from a vehicle monitored state to a vehicle non-monitored state, instructions executed by computer **108** can suppress display of parameters related to an internal operating environment of vehicle **102**, such as sensors **107** to sense vehicle speed, engine torque, braking sensors, temperatures of components connected or coupled to actuators **110**, etc.

[0044] Based on placement of vehicle **102** in a non-monitored state, instructions executed by computer **108** can transition display **104** to a content delivery state, in which content generated by server **118** can be displayed. In an example, server **118** can execute instructions to implement a videoconferencing bridge. In an example, a videoconferencing bridge may include a media server, a control unit, and a network interface. The control unit of server **118** may operate to track destination addresses of videoconference participants as well as tracking packets transmitted during the videoconference to determine whether transmitted packets reach an intended destination. The network interface of server **118** may operate to provide connectivity among videoconference participants and to determine whether transmitted packets arrive at an address of a conference participant in an intended order. The media server of server **118** may operate to encode images transmitted during the videoconference and to decode images received during the videoconference.

[0045] In an example, in a content delivery state, images (e.g., real-time images) of videoconference participants, as well as content provided by a videoconference participant (e.g., view graphs that include still or video images) may be displayed in place of images of a vehicle's external environment and/or images relating to a vehicle's internal operating state. In an example, display **104** can include multiple displays (e.g., **104**A and **104**B of FIGS. **2** and **3**) to display images of videoconference participants (as further described in reference to FIG. **3**) while camera **127** (of FIG. **2**) captures still or moving images of the operator of vehicle **102**. In an example, still or moving images of the operator of vehicle **102** can be processed by a videoconferencing bridge at server **118** to introduce blurring or removal of items in the background of the operator of vehicle **102**, such as passengers positioned in the front or rear seats of vehicle **102**, cargo secured within the interior of vehicle **102**, etc. Alternatively or in addition, in a content delivery state, vehicle computer **108** can be trained during a calibration process to recognize cargo or other images aside from the operator of vehicle **102** so as to (without human input) recognize images of the interior of vehicle **102** and two blur or remove such images from video transmitted from the vehicle to server **118**.

[0046] Audio from conference participants can be presented to the operator via an infotainment subsystem of vehicle **102**. While display **104** operates in a content delivery state, haptic interfaces of HMI **112** can be utilized to provide outputs to the operator, such as via a haptic actuator on a steering wheel of vehicle **102**. In one example, at the start of a videoconference, a haptic actuator positioned on a steering wheel of vehicle **102** can provide an output to the operator to take notice of the initiation of the videoconference. In another example, a haptic actuator positioned on the driver side seat may provide an output to the operator to take notice of content that may be of particular interest to the operator, such as posting of a videoconference text message. In an example, a videoconference participant may indicate that a particular portion of presented content may be of particular interest to the operator. In such an example, the videoconference participant may provide an output to the operator, which may be converted (e.g., via instructions executed by computer **108**) to an activation of a steering wheel mounted or seat mounted haptic actuator.

Exemplary System Operations

[0047] FIG. **2** is a diagram **200** showing displays positioned in an interior portion of vehicle **102**. In the example of FIG. **2**, vehicle **102** operates in a monitored state, in which an externally facing camera captures images of static or moving vehicles in the traffic environment of vehicle **102**. Display **104**A, which can include a panoramic display that extends across a dashboard of vehicle **102** from a first location proximate to a left interior boundary to a second location proximate to a right interior boundary. Display **104**A can display information such as vehicle speed, vehicle direction of travel, operator alerts, etc. Display **104**B may provide a touchscreen capability, which permits an operator of vehicle **102** to control settings of an onboard infotainment system (e.g., radio settings), climate control settings, seat positions, headlamp illumination, door lock/unlock status, etc. In the example of FIG. **2**, images of vehicles operating in the traffic environment of vehicle **102** can be augmented utilizing inputs from one or more of sensors **107** (e.g., radar, lidar, etc.) to classify whether an object is static (e.g., a parked car) or moving (e.g., in traffic), so as to provide an operator of vehicle **102** with situational awareness of the status of objects/obstacles in the path of travel of vehicle **102**. In an example, display **104**B may display parameters related to operations internal to vehicle **102**, such as an engine temperature, a time to a destination, a remaining fuel quantity or a battery state of charge, etc.

[0048] In the example of FIG. **2**, camera **127** can be utilized to capture an image of an individual seated in the driver position of vehicle **102**. After capturing an image of an individual seated in the driver position of vehicle **102**, the computer **108** can execute instructions to determine whether an individual's facial and/or other features represent those of an authorized operator of vehicle **102**. Accordingly, camera **127** interacting with computer **108** can be utilized in a facial recognition process to biometrically identify the individual, thereby allowing the individual to operate the vehicle and/or components thereof. Operation of the vehicle or vehicle components may include starting the vehicle, controlling propulsion of the vehicle, steering the vehicle, accessing HMI **112**, etc. In a facial recognition process, features of the individual's face and/or other portions of the individual may be compared with a set of stored facial and/or other parameters that permit computer **108** to identify an operator and to determine whether the operator is authorized to operate the vehicle and/or access or carry out other operations in the vehicle.

[0049] Microphone **125** can receive voice inputs and/or voice commands from an operator of vehicle **102**. Accordingly, in one example, microphone **125** can receive a voice command, such as a command to start the engine of vehicle **102**. Output signals from microphone **125** can be transmitted to HMI **112**, which can process signals from microphone **125** and transmit a suitably formatted command (e.g., an engine start command) to computer **108**. In an example, microphone **125** can be a directional microphone with a processor and memory that includes a capability to determine a direction from which an audio command emanates. Accordingly, in an example, microphone **125** is capable of determining whether an audio or voice signal emanates from an operator of vehicle **102**, a passenger positioned at the right hand side of the driver, a passenger in a rear seat of vehicle **102**, etc.

[0050] FIG. **3** is a diagram **300** showing displays positioned at in an interior portion of vehicle **102**. In the example of FIG. **3**, vehicle **102** operates in a non-monitored state, in which images captured by an externally facing camera are suppressed. Vehicle **102** may operate in a non-monitored state in response to an operator of the vehicle exiting a driving environment (e.g., positioning the vehicle at a parking lot) and placing the transmission of vehicle **102** in a state (e.g., a "park" state) that prevents movement of the wheels of the vehicle. Based on placement of vehicle **102** in a non-monitored state, instructions executed by computer **108** may transition vehicle displays **104**A and **104**B to a content delivery state. In a content delivery state, displays **104**A and **104**B may be capable of receiving videoconferencing transmissions from server **118**, such as images of videoconference participants, content presented by videoconference participants (e.g., view graph materials, charts, multimedia segments, etc.).

[0051] In the example of FIG. **3**, camera **127** can be utilized to capture an image of an individual seated in the driver position of vehicle **102** during a videoconferencing session. After capturing a still or moving image of an individual seated in the driver position of vehicle **102**, instructions executed by computer **108** can operate to transmit the still or moving image of the individual to server **118**. Accordingly, camera **127** interacting with computer **108**, can be utilized to permit the operator of vehicle **102** to participate in a videoconferencing session. In an example, during operation of vehicle **102** in a non-monitored state, certain facial recognition processes, such as processes to permit an individual to start the vehicle, control propulsion of the vehicle, steer the vehicle, etc., can be suppressed. In an example, in a non-monitored state, a facial recognition process can be utilized to identify an operator of vehicle **102**. Such identification can permit instructions executing on a videoconference bridge of server **118** to post the name of the operator or of another individual seated in the interior of vehicle **102**.

[0052] In an example, in a non-monitored state, voice commands from the operator of vehicle **102** may also be suppressed. Accordingly, in an example, an audio signal processing capability of HMI **112**, which operates to process signals from microphone **125** and transmit suitably formatted commands to HMI **112** (e.g., an engine start command) can be suppressed. Thus, audio signals from the operator may be transmitted via communications component **114** to server **118** without further processing by HMI **112** and/or computer **108**. In an example, microphone **125** includes a directional microphone capable of determining a direction from which audio signals emanate. Accordingly, in an example, HMI **112** can operate to filter audio signals received by microphone **125** so as to transmit audio that exclusively emanates from the operator. Thus, during a videoconferencing session, audio signals emanating from passengers located within the interior of vehicle **102**, who may be conversing between or among each other, can be deemphasized or removed from an audio stream transmitted to server **118**. In another example, during a videoconferencing session, audio signals emanating from an individual positioned in the passenger seat can be emphasized (or amplified) while audio signals emanating from an individual positioned in the driver seat can be deemphasized or removed.

[0053] While vehicle **102** operates in a non-monitored state, HMI **112** can generate haptic outputs to an operator, such as activating a haptic actuator located in the driver side seat or on the steering wheel. In an example, a haptic actuator may output a predetermined series of vibrations, e.g., a vibration at a specified intensity for a specified period of time, which may notify the operator of vehicle **102** of the initiation of a videoconferencing session. In another example, a haptic actuator can notify the operator of vehicle **102** that a potential participant has requested access to the videoconference. In another example, a haptic actuator may output of predetermined series of vibrations to notify the operator of vehicle **102** that a videoconference participant has indicated that certain content is relevant to the operator. In another example, a haptic actuator located in the driver side seat of vehicle **102** may notify the operator of vehicle **102** that a text message has been posted during a videoconference session, that a potential participant is seeking entry into the videoconference session, that a participant has left the videoconference session, etc.

[0054] FIG. **4** is a block diagram **400** showing vehicle operation of a vehicle system in a vehicle monitored state and in a vehicle non-monitored state. As illustrated in FIG. **4**, computer **108**, HMI **112**, and displays **104**A/**104**B are capable of operating in states representing vehicle monitored and vehicle non-monitored states. In an example, in a vehicle monitored state, computer **108** can execute vehicle monitoring state component **408**A to monitor the environment external to vehicle **102** utilizing output signals from suitable sensors of sensors **107**. Such sensors can include cameras, radar sensors, lidar sensors, etc. In an example, vehicle monitoring state component **408**A additionally includes instructions to monitor the environment internal to vehicle **102** utilizing output signals from suitable sensors of sensors **107**, which can include fuel level sensors, engine temperature sensors, wheel speed sensors, oil pressure and temperature sensors, etc.

[0055] In a vehicle monitored state, a processor coupled to a memory of HMI **112** can execute vehicle monitoring state component **412**A to receive audio commands (e.g., via microphone **125**) from an operator of vehicle **102** and to provide audio and/or haptic outputs to the operator. In an example, haptic outputs may notify the operator of an approaching (second) vehicle. In another example, a haptic actuator may output a predetermined series of vibrations to notify the operator of vehicle **102** of upcoming traffic congestion along path **50** (of FIG. **1**). Such haptic outputs can include a vibration actuator in a driver side seat of vehicle **102**, a vibration actuator in a steering wheel of the vehicle, or another haptic or tactile actuator located in an interior of the vehicle. In an example, in a vehicle monitored state, display **104**A and/or display **104**B can operate in vehicle monitoring state **404**A to display static or moving objects viewable by an externally facing camera as well as to provide visual indications of the status of internal operating parameters of vehicle **102**.

[0056] In the example of FIG. **4**, based on the operator setting or placing transmission switch **405** of vehicle **102** into a state that prevents movement or rotation of the wheels of the vehicle, computer **108** can receive a signal from transmission switch **405** and transition the vehicle from a monitored state to a non-monitored state. In a vehicle non-monitored state, vehicle non-monitoring state component **408**B can operate to suppress monitoring of the environment external to vehicle **102** and to suppress monitoring of an operating environment internal to the vehicle. In an example, based on the vehicle transitioning to the non-monitored state, computer **108** can execute instructions to output a signal to HMI **112** so as to instruct HMI **112** to execute instructions of vehicle non-monitoring state component **412**B. In addition, based on the vehicle transitioning to the non-monitored state, computer **108** can execute instructions to output a signal to display **104**A and/or display **104**B so as to instruct a display to initiate content delivery state **404**B.

[0057] In an example, based on receipt of an output signal from computer **108**, HMI **112** can execute instructions of vehicle non-monitoring state component **412**B. Instructions of vehicle non-monitoring state component **412**B can operate to suppress receipt of audio commands from an operator of vehicle **102**. In addition, component **412**B can include instructions to suppress audio and/or haptic outputs relating to the environment external to the vehicle or to an operating environment internal to vehicle **102**. In an example, non-monitoring state component **412**B can operate to disable volume controls of HMI **112** that relate to an onboard infotainment system, a satellite radio system, etc., and permit such controls to modify a volume setting for audio from a videoconferencing participant. In an example, HMI **112** non-monitoring state component **412**B can implement a plurality of volume controls, utilizing, for example, display **104**B. Thus, in an example, the operator may be permitted to adjust (e.g., increase or decrease) a volume setting with respect to any selected participant present during the videoconferencing session.

[0058] In an example, non-monitoring state component **412**B can include capability to display a scene captured by camera **127**, which may permit an operator or passenger of vehicle **102** to select the individuals, e.g., via a touch screen of display **104**B, that are to be visible to other participants of videoconference session. Alternatively or in addition, a touchscreen of display **104**B may permit an operator to select items in the background of a scene captured by camera **127** that are to be blurred or removed from a video stream transmitted to other videoconference participants. Accordingly, in an example, an operator can select that a passenger be visible to other participants of a videoconference while blurring or replacing images of other passengers of vehicle **102**, such as passengers positioned in a rear seat of the vehicle. In another example an operator can select an image of secured cargo, such as cargo positioned in a rear seat of vehicle **102**, for blurring or removal from video transmitted to other videoconference participants. In an example, images removed from video transmitted to other video conference participants can be replaced by a background image, such as an image of an office environment or other setting.

[0059] Alternatively or in addition, vehicle computer **108** can be trained during a calibration process to recognize nonparticipant passengers, cargo, and/or other images present in the field of view of camera **127**. In an example, during a calibration process, an off-line neural network can be trained so as to generate one or more parameters for uploading into a memory of computer **108**. A parameter can permit instructions executed by computer **108** to recognize (e.g., without human input) images in the field of view of camera **127** that are aside from the operator of vehicle **102**, such as images of passengers positioned in a rear seat or in the passenger seat of vehicle **102**, secured cargo in the field of view of camera **127**, etc., and to blur or remove such images from video transmitted to server **118**. For example, during a calibration process, an off-line neural network, such as a convolutional neural network having at least three layers (i.e., an input layer, an output layer, and at least one hidden layer) can be utilized. In an example, the input layer of the neural network can operate to receive an image of an interior portion of vehicle **102**, which includes images of an operator along with additional images, such as images that represent passengers, secured cargo, or other items aside from the operator. A loss function can then be computed to express a discrepancy between actual content of a scene of the interior of vehicle **102** and a decision presented at the output layer of the neural network. Via supervised learning, semi-supervised learning, reinforcement learning, etc., weighting functions of the hidden layer of the neural network can be adjusted to reduce one or more components of the loss function, thereby increasing the likelihood of the neural network correctly distinguishing an operator of vehicle **102** from passengers within the interior of vehicle **102**. After the neural network has been suitably trained, weighting functions can be transformed into one or more parameters for uploading to computer **108**. Accordingly, during operation of displays **104**A and/or **104**B in a content delivery state, computer **108** (operating in a vehicle non-monitoring state) can, without human input, blur or remove images aside from images corresponding to the operator of vehicle **102**.

[0060] Vehicle non-monitoring state component **412**B can additionally include instructions to actuate audio and/or haptic outputs to the operator that relate or pertain to, for example, activities related to videoconferencing capabilities. Accordingly, in response to instructions executed by a processor of server **118**, HMI **112** can activate a vibration actuator in a driver side seat of vehicle **102**, a vibration actuator in the steering wheel of vehicle **102**, etc. In an example, in response to a potential videoconference participant seeking entry to an ongoing videoconference, vehicle non-monitoring state component **412**B can activate an actuator in a driver side seat so as to provide the opportunity for the operator to admit the potential participant to the videoconference. In another example, in response to instructions executed by a processor of server **118** determining that content presented by a videoconference participant is of relevance to the operator of vehicle **102**, HMI **112** can activate a vibration actuator and/or an audio output.

[0061] In another example, based on an operator appearing to be disengaged from a videoconference, vehicle non-monitoring state component **412**B may interact with vehicle non-monitoring state component **408**B to detect reorientation of head or facial features of an operator of the vehicle. Accordingly, in an example, in response to an image captured via a camera **127** indicating that the operator's head or face appears to be directed away from one or more of displays **104**A and/or **104**B (e.g., looking out of a side window of vehicle **102**) vehicle non-monitoring state component **412**B can activate a haptic actuator and/or an audio output to the operator to take notice of an ongoing videoconference.

[0062] In an example, at the conclusion of a videoconference, computer **108**, HMI **112**, and one or more of displays **104**A and **104**B can return to a vehicle monitoring state (e.g., **408**A, **412**A, and **404**A). A return to a monitoring state can be triggered in accordance with vehicle non-monitoring state component **408**B of computer **108** receiving a signal from server **118**. Alternatively or in addition, a return to a monitoring state can be triggered by vehicle non-monitoring state component **408**B detecting a change of state of transmission switch **405**. The change of state of transmission switch **405** can be based on an operator placing or setting a transmission switch **405** of vehicle **102** into a state that permits movement or rotation of the wheels of vehicle **102**.

[0063] FIG. **5** is a flowchart of an example method or process **500** for controlling a display **104**, including transitioning between a vehicle monitored state and a vehicle non-monitored state. As a general overview, process **500** can include an operator placing the vehicle into a non-monitored state, such as by placing the transmission of vehicle **102** in "park," or another state in which the transmission of the vehicle prevents movement of the wheels. Based on vehicle **102** being placed in a non-monitored state, vehicle displays **104**A and/or **104**B can transition to a content delivery state. In a content delivery state, display of an environment external to vehicle **102** and display of parameters relating to operations internal to the vehicle are suppressed. In the content delivery state, a videoconference or other content delivery activity (e.g., real-time videoconferencing, video telephony, etc.) can be initiated. During operation in a content delivery state, vehicle displays **104**A and/or **104**B can operate to present images of videoconference participants and content shared by videoconference participants. In addition, during operation in a content delivery state, HMI **112** can operate to provide haptic and/or audio alerts in the form of audio signaling, activation of a vibration actuator in a driver's seat or in a steering wheel, etc. Further, during operation in a content delivery state, vehicle infotainment controls, radio settings, etc., may be disabled to permit an operator to adjust volume from a videoconference participant. In response to the vehicle being placed into a monitored state, such as by placing the transmission of vehicle **102** in "drive," "reverse," or other setting that permits movement of the wheels of the vehicle, the vehicle may return to a monitored state. In a monitored state, presentation of videoconferencing content can be suppressed so that images of an environment external to vehicle **102** and parameters related to internal operations of the vehicle can be displayed. In an example, after placing the transmission of vehicle **102** into a setting that permits movement of the wheels of the vehicle, HMI **112** can maintain audio communications with videoconference participants so as to permit the operator of vehicle **102** to continue communicating with videoconference participants while operating vehicle **102** in a traffic environment.

[0064] Process **500** begins at block **505**, which includes determining that vehicle **102** has been set or placed in a non-monitored state. In an example, computer **108** may determine that vehicle **102** has been placed in a non-monitored state by detecting transmission switch **405** of the vehicle being set or placed in a "park" state or in another state which the transmission of the vehicle prevents movement of the wheels.

[0065] Process **500** continues at block **510**, which includes computer **108** of vehicle **102** transitioning the vehicle display and control from a monitoring state to a non-monitoring state. In a non-monitoring state, images, symbols, and other graphics that display information relative to a detected object external to vehicle **102** can be suppressed. In addition, in a non-monitoring state, display of images related to the operating environment internal to vehicle **102**, such as display of parameters related to engine speed, engine and transmission temperatures, battery temperatures, etc., can also be suppressed. In an example, block **510** can include computer **108** transitioning from executing instructions of vehicle monitoring state component **408**A to executing instructions of vehicle non-monitoring state component **408**B. In an example, block **510** can include HMI **112** transitioning from executing instructions of vehicle monitoring state component **412**A to vehicle non-monitoring state component **412**B.

[0066] Process **500** continues at block **515**, which includes computer **108** of vehicle **102** transitioning displays **104**A and/or **104**B from vehicle monitoring state **404**A to content delivery state **404**B. In a content delivery state, content from an external server that implements, e.g., a videoconference bridge, can be displayed utilizing one or more of the transitioned displays. In response to initiating a content delivery state, display **104**A can display images of videoconference participants, content (e.g., charts, view graphs, still or video images, etc.) can be displayed using, e.g., a panoramic display of the vehicle that extends across the dashboard of the vehicle from a first location proximate to a left interior boundary of the interior of vehicle **102** to a second location proximate to a right interior boundary of the vehicle. Alternatively or in addition, display **104**B, which may be positioned beneath dashboard **120** of vehicle **102**, can display additional videoconference related information such as a listing of participants, time remaining in a videoconference, a videoconference title, etc. In an example, microphone **125** can be used to receive audio from the operator of vehicle **102** and transmit the received audio to videoconference participants as directed by a videoconferencing bridge executed via server **118**. Audio from videoconference participants can be transmitted to the operator via an audio system located in the vehicle's interior. In in content delivery state **404**B, vehicle computer **108** can utilize one or more parameters to blur or remove images of passengers positioned in the passenger seat of vehicle **102**, passengers positioned in the rear of vehicle **102**, and/or cargo secured within the interior of vehicle **102**. Such blurring or removal of images can occur responsive to an operator of vehicle **102**, for example, selecting to remove certain images from video transmitted from vehicle **102**. Alternatively or in addition, blurring or removal of images can occur without operator input, such as via off-line training of a neural network, which may result in an upload of one or more parameters to vehicle computer **108**.

[0067] Process **500** continues at block **520**, which includes operating vehicle display and control functions in a content delivery state. In an example, non-monitoring state component **412**B can operate to disable volume controls of HMI **112** that relate to an onboard infotainment system, a satellite radio system, etc., and permit such controls to modify a volume setting for audio from a videoconferencing participant. In an example, HMI **112** non-monitoring state component **412**B can implement a plurality of volume controls, utilizing, for example, display **104**B. In an example, the operator may be permitted to adjust (e.g., increase or decrease) a volume setting from any selected participant present during the videoconferencing session. Block **520** can additionally include HMI **112** activating a haptic actuator and/or an audio output to the operator to take notice of content presented during the videoconference.

[0068] Process **500** continues at block **525**, which includes determining that the vehicle has transitioned to a monitored state. In an example, a transition to a monitored state can be based on a detected change in the state of transmission switch **405**, which can be transitioned from a state that prevents movement of the wheels of the vehicle to a state that permits movement of the wheels.

[0069] Process **500** continues at block **530**, which includes, in response to detection of the change of state of transmission switch **405**, transitioning of vehicle displays **104**A and/or **104**B from a content delivery state to a state that permits display of aspects of an environment external to vehicle **102** and aspects of an operating environment internal to the vehicle. In an example, block **530** can include computer **108** transitioning from executing instructions of vehicle non-monitoring state component **408**B to executing instructions of vehicle monitoring state component **408**A. In an example, block **530** can include HMI **112** transitioning from executing instructions of vehicle non-monitoring state component **412**B to vehicle monitoring state component **412**A. In an example, block **530** can include displays **104**A and/or **104**B transitioning from content delivery state **404**B to vehicle monitoring state **404**A. In an example, block **530** can include HMI **112** maintaining audio communications with videoconference participants so as to permit the operator of vehicle **102** to continue communicating with videoconference participants while operating vehicle **102** in a traffic environment.

[0070] After completing block **530**, process **500** ends.

[0071] Operations, systems, and methods described herein should always be implemented and/or performed in accordance with an applicable owner's/user's manual and/or safety guidelines.

[0072] In general, the computing systems and/or devices described may employ any of a number of computer operating systems, including, but by no means limited to, versions and/or varieties of the Ford Sync® application, AppLink/Smart Device Link middleware, the Microsoft Automotive® operating system, the Microsoft Windows® operating system, the Unix operating system (e.g., the Solaris® operating system distributed by Oracle Corporation of Redwood Shores, California), the AIX UNIX operating system distributed by International Business Machines of Armonk, New York, the Linux operating system, the Mac OSX and iOS operating systems distributed by Apple Inc. of Cupertino, California, the BlackBerry OS distributed by Blackberry, Ltd. of Waterloo, Canada, and the Android operating system developed by Google, Inc. and the Open Handset Alliance, or the QNX® CAR Platform for Infotainment offered by QNX Software Systems. Examples of computing devices include, without limitation, an onboard vehicle computer, a computer workstation, a server, a desktop, notebook, laptop, or handheld computer, or some other computing system and/or device.

[0073] Computing devices generally include computer-executable instructions, where the instructions may be executable by one or more computing devices such as those listed above. Computer executable instructions may be compiled or interpreted from computer programs created using a variety of programming languages and/or technologies, including, without limitation, and either alone or in combination, Java™, C, C++, Matlab, Simulink, Stateflow, Visual Basic, Java Script, Python, Perl, HTML, etc. Some of these applications may be compiled and executed on a virtual machine, such as the Java Virtual Machine, the Dalvik virtual machine, or the like. In general, a processor (e.g., a microprocessor) receives instructions, e.g., from a memory, a computer readable medium, etc., and executes these instructions, thereby performing one or more processes, including one or more of the processes described herein. Such instructions and other data may be stored and transmitted using a variety of computer readable media. A file in a computing device is generally a collection of data stored on a computer readable medium, such as a storage medium, a random access memory, etc.

[0074] A computer-readable medium (also referred to as a processor-readable medium) includes any non-transitory (e.g., tangible) medium that participates in providing data (e.g., instructions) that may be read by a computer (e.g., by a processor of a computer). Such a medium may take many forms, including, but not limited to, non-volatile media and volatile media. Instructions may be transmitted by one or more transmission media, including fiber optics, wires, wireless communication, including the internals that comprise a system bus coupled to a processor of a computer. Common forms of computer-readable media include, for example, RAM, a PROM, an EPROM, a FLASH-EEPROM, any other memory chip or cartridge, or any other medium from which a computer can read.

[0075] Databases, data repositories or other data stores described herein may include various kinds of mechanisms for storing, accessing, and retrieving various kinds of data, including a hierarchical database, a set of files in a file system, an application database in a proprietary format, a relational database management system (RDBMS), a non-relational database (NoSQL), a graph database (GDB), etc. Each such data store is generally included within a computing device employing a computer operating system such as one of those mentioned above, and are accessed via a network in any one or more of a variety of manners. A file system may be accessible from a computer operating system, and may include files stored in various formats. An RDBMS generally employs the Structured Query Language (SQL) in addition to a language for creating, storing, editing, and executing stored procedures, such as the PL/SQL language mentioned above.

[0076] In some examples, system elements may be implemented as computer-readable instructions (e.g., software) on one or more computing devices (e.g., servers, personal computers, etc.), stored on computer readable media associated therewith (e.g., disks, memories, etc.). A computer program product may comprise such instructions stored on computer readable media for carrying out the functions described herein.

[0077] In the drawings, the same reference numbers indicate the same elements. Further, some or all of these elements could be changed. With regard to the media, processes, systems, methods, heuristics, etc. described herein, it should be understood that, although the steps of such processes, etc. have been described as occurring according to a certain ordered sequence, such processes could be practiced with the described steps performed in an order other than the order described herein. It should further be understood that certain steps could be performed simultaneously, that other steps could be added, or that certain steps described herein could be omitted.

[0078] All terms used in the claims are intended to be given their plain and ordinary meanings as understood by those skilled in the art unless an explicit indication to the contrary in made herein. In particular, use of the singular articles such as "a," "the," "said," etc. should be read to recite one or more of the indicated elements unless a claim recites an explicit limitation to the contrary. The adjectives "first" and "second" are used throughout this document as identifiers and are not intended to signify importance, order, or quantity. Use of "in response to" and "upon determining" indicates a causal relationship, not merely a temporal relationship.

[0079] The disclosure has been described in an illustrative manner, and it is to be understood that the terminology which has been used is intended to be in the nature of words of description rather than of limitation. Many modifications and variations of the present disclosure are possible in light of the above teachings, and the disclosure may be practiced otherwise than as specifically described.

What is claimed is:

1. A system, comprising a computer including a processor and memory, the memory storing instructions executable by the processor to:

transition a vehicle display from a vehicle monitored state to a content delivery state based on a placement of the vehicle in a non-monitored state; and

transition a control from a vehicle control state to a content delivery control state based on the placement of the vehicle in the non-monitored state.

2. The system of claim **1**, wherein the non-monitored state includes a state of a vehicle transmission switch.

3. The system of claim **1**, wherein the vehicle display extends across a dashboard of the vehicle from a first location proximate to a left interior boundary to a second location proximate to a right interior boundary.

4. The system of claim **1**, wherein the instructions to transition the vehicle display include instructions to suppress display of images that represent an environment external to the vehicle.

5. The system of claim **1**, wherein the instructions to transition the vehicle display include instructions to suppress display of operating parameters of a vehicle system.

6. The system of claim **1**, wherein the instructions to transition the control from the vehicle control state include instructions to filter audio signals emanating from an interior portion of the vehicle to include audio exclusively emanating from an operator of the vehicle or from a passenger of the vehicle.

7. The system of claim **1**, wherein the content delivery state is a videoconferencing state, the videoconferencing state permitting display of content transmitted from a source external to the vehicle.

8. The system of claim **7**, wherein the instructions to transition the control from the vehicle control state include instructions to modify a volume setting for audio directed to a videoconferencing participant.

9. The system of claim **7**, wherein the instructions to transition the control from the vehicle control state include instructions to detect reorientation of head or facial features of an operator of the vehicle.

10. The system of claim **7**, wherein the instructions to transition the control from the vehicle control state include instructions to generate an audio output responsive to determining that videoconferencing content pertains to an operator of the vehicle.

11. The system of claim **7**, wherein the instructions to transition the control from the vehicle control state include instructions to generate a haptic output to an operator of the vehicle responsive to determining audio or video content that pertains to the operator of the vehicle.

12. The system of claim **7**, wherein the instructions to transition the control from the vehicle control state include instructions to suppress video signals from an operator of the vehicle based on detecting that the operator of the vehicle has become disengaged from the videoconference.

13. The system of claim **7**, wherein the instructions to transition the control from the vehicle control state include instructions to suppress a video portion of a videoconference responsive to a determination that the vehicle has been placed in a monitored state.

14. The system of claim **7**, wherein the instructions to transition the control from the vehicle control state include instructions to maintain an audio portion of a videoconference responsive to a determination that the vehicle has been placed in a monitored state.

15. A method, comprising:

transitioning a vehicle display from a vehicle monitored state to a content delivery state based on a placement of the vehicle in a non-monitored state; and

transitioning a control from a vehicle control state to a content delivery control state based on the placement of the vehicle in the non-monitored state.

16. The method of claim **15**, wherein determining that the vehicle has been placed in the non-monitored state comprises determining a state of a vehicle transmission switch.

17. The method of claim **15**, wherein the vehicle display extends across a dashboard of the vehicle from a first location proximate to a left interior boundary to a second location proximate to a right interior boundary.

18. The method of claim **15**, wherein transitioning the vehicle display includes suppressing display of images representing an environment external to the vehicle.

19. The method of claim **15**, wherein transitioning the vehicle display includes suppressing display of operating parameters of a vehicle system.

20. The method of claim **15**, wherein the content delivery state is a videoconferencing state, the videoconferencing state permitting display of content transmitted from a source external to the vehicle.

* * * * *